KRITHIKA

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Software Engineer with 3+ years building production ML infrastructure, distributed training systems, and generative AI pipelines at scale. Experienced in model deployment, large-scale data processing, and ML system reliability.

EXPERIENCE

EOG Resources | Data Engineer II 2023 – Present

ML Infrastructure & Distributed Systems

- Built a Kubernetes + Dagster self-service ML platform enabling 10+ Reservoir Engineers to independently validate models, cutting Data Science review cycles by 40%.
- Developed a Slurm CLI tool for Data Scientists to submit and monitor distributed training jobs; configured multi-node GPU topology reducing model training time by 35%.
- Deployed FastAPI inference services handling 5+ team workloads with sub-second response times at high concurrency; standardized experiment tracking and deployment via MLflow, accelerating release velocity by 30%.

Generative AI & LLM Systems

- Architected PGVector semantic search over unstructured document corpora, reducing retrieval latency by 60%; built Amazon Textract ingestion pipelines for Llama and Mistral fine-tuning, processing 50K+ documents at scale.

VLM Inference Pipeline

- Built a vision-language model inference pipeline using Gemini VLM and DSPy prompt optimization to detect oil pipeline leaks from aerial and ground imagery, replacing manual inspection workflows.
- Designed DSPy-based prompt optimization loop to iteratively improve leak detection precision; integrated inference service into existing IoT monitoring infrastructure for real-time image-triggered alerting.

Data Platform Engineering

- Engineered IoT ingestion pipelines for 500+ industrial sensors, processing 7,500+ data tags per minute company-wide; built a digital twin audio pipeline for real-time equipment health monitoring with rule-based anomaly detection.
- Architected RabbitMQ event-driven messaging layer increasing system throughput by 25%; implemented Datadog telemetry and alerting across all pipelines, reducing system downtime by 40%.

Regional Gallopt *(Reactive Control System)*

- Designed a low-latency hydraulic control model monitoring oil sales point pressure across 200+ wells, triggering automated decisions (choking, injection reduction, shut-ins) within milliseconds; achieved zero safety incidents post-deployment.

EOG Resources | Data Engineer Intern May – August 2022

- Computed 3D tortuosity indices across 3,000+ well sites in Python, building a regression model correlating borehole deviation with production output and surfacing actionable drilling insights.
- Developed an interactive geospatial dashboard used by 1,000+ engineers for spatial coverage analysis, replacing ad hoc manual reporting with low-latency data products.

TECHNICAL SKILLS

Cloud & Orchestration: Kubernetes (CKAD), Slurm, Docker, Rancher, GitHub Actions, Terraform
Workflow & Messaging: Apache Airflow, Dagster, RabbitMQ, MQTT, Pulsar, Azure Event Hub
Data Systems: PostgreSQL (PGVector), Oracle, Amazon S3, Apache NiFi, Apache Spark
Generative AI & MLOps: MLflow, LLM fine-tuning (Llama, Mistral), Gemini VLM, DSPy, Amazon Textract, semantic search
Languages & Monitoring: Python, TypeScript, SQL | Datadog, Grafana, Elastic APM

EDUCATION & CERTIFICATIONS

M.S. Computer Science | University of Houston GPA: 3.8/4.0
M.S. Software Systems | Coimbatore Institute of Technology GPA: 9.0/10
Certified Kubernetes Application Developer (CKAD) | Linux Foundation

PUBLICATIONS

Leveraging Marker-based AR for Just-in-time Safety Information Presentation (ASEE 2022) | Ensuring Data Integrity in Cloud Storage (ICCCNT 2018)